

Ardelmax — Stability Study Design

Protocol STAB-P-2024-53

Sample Inventory and Pull Windows

Sample inventory per condition and timepoint is defined in the pull schedule appendix. Scheduled pulls are performed within ± 7 days of the nominal date; pull records are maintained in the site stability management system.

Data Evaluation and Reporting

Stability data are evaluated per ICH Q1E. Regression analysis is applied to quantitative attributes exhibiting change over time; poolability across batches is assessed at the 0.25 significance level. Annual reports summarize all data generated in the reporting interval.

Study Storage Conditions

The primary (intermediate/zone IVa condition) storage condition for this protocol is $30^{\circ}\text{C} \pm 2^{\circ}\text{C}$ / 65% RH $\pm 5\%$ RH. Accelerated studies at $40^{\circ}\text{C} \pm 2^{\circ}\text{C}$ / 75% RH $\pm 5\%$ RH were completed through six months during development and are not part of this protocol.

Pull Schedule

Long-term samples are tested at 0, 3, 6, 9, 12, 18, 24 and 36 months.

Deviations

Chamber excursion of $+3^{\circ}\text{C}$ for 12 hours affected the 12-month interval for batch 002; mean kinetic temperature remained within limits.

Acceptance Criteria

The stability acceptance criterion applied at each interval is — Water content: not more than 5.0% w/w. All other registered specification tests apply unchanged.

Bracketing and Matrixing

A bracketing design per ICH Q1D tests the extremes of the strength range (20 mg and 40 mg); the 25 mg strength shares identical formulation composition and packaging and is therefore not tested.

Stability Commitments

As a condition of approval, three consecutive production batches of Ardelmax will be entered into the stability program, and the retest period will be re-evaluated after 36-month data are available.

Container Closure and Orientation

Samples are stored in the commercial container closure system. Bottles are stored upright; a subset of inverted samples is included at the initial and final timepoints to assess closure-liner interaction. Induction seal integrity is verified at each pull.

Data Evaluation and Reporting

Stability data are evaluated per ICH Q1E. Regression analysis is applied to quantitative attributes exhibiting change over time; poolability across batches is assessed at the 0.25 significance level. Annual reports summarize all data generated in the reporting interval.

Batch Selection

The protocol covers three primary batches manufactured at commercial scale using the approved process. Batch genealogy, manufacturing dates, and packaging lots are recorded in the batch appendix. All batches were released against the approved specification prior to set-down.

Analytical Procedures

Assay and degradation products are determined by the stability-indicating reversed-phase HPLC procedure described in the analytical methods dossier. Dissolution testing uses USP Apparatus II at 50 rpm. All procedures were validated in accordance with ICH Q2(R1) prior to protocol initiation.